

Etalé space & Sheafification

Let $\mathcal{F}$ be a presheaf on some topological space X .

Consider the set $\overline{\mathcal{F}} := \coprod_{x \in X} \mathcal{F}_x$, where $\mathcal{F}_x$ denotes the stalk of $\mathcal{F}$ over x .

Then there is a natural projection map $\pi : \overline{\mathcal{F}} \rightarrow X$ given by $\mathcal{F}_x \mapsto x$.

We define the étalé space as a topology on $\overline{\mathcal{F}}$ as follows:

Suppose that U is open in X such that $s \in \mathcal{F}(U)$.

Then we can construct the function $\overline{s} : U \rightarrow \overline{\mathcal{F}}$ given by $x \mapsto [s]_x$, which is a section of the projection map π on U .

We define a topology on $\overline{\mathcal{F}}$ with basis given by sets of the form $\overline{s}(U)$.

The sheafification of $\mathcal{F}$, henceforth denoted $\tilde{\mathcal{F}}$, is the sheaf of continuous sections of the projection map π .

In other words, $\tilde{\mathcal{F}}(U) = \{f : U \rightarrow \overline{\mathcal{F}} \mid f \text{ is a continuous section of } \pi\}$.

We give an alternate characterization of what it means for f to be a continuous section of π .

$f \in \tilde{\mathcal{F}}(U)$ is continuous

$\iff f^{-1}(V)$ is open for all open $V \subseteq \overline{\mathcal{F}}$

$\iff f^{-1}(\overline{s}(W))$ is open for all open $W \subseteq X$ and s

$\iff f^{-1}(\overline{s}(W)) = \{x \in W \mid f(x) = [s]_x\}$ contains a neighbourhood of all of its points

$\iff$ for all $y \in f^{-1}(\overline{s}(W))$, there exists an open neighbourhood W_y of y in W and a section $s \in \mathcal{F}(W_y)$ such that $f|_{W_y} = \overline{s}|_{W_y}$.

The fact that f is a π -section guarantees that the image of any point $x \in U$ has at most one element in $\mathcal{F}_x$; i.e. the information of any point is not "doubled" in $\mathcal{F}$.